Project Euler Problem 7: 10 001st Prime

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Problem

By listing the first six prime numbers: $2, 3, 5, 7, 11$, and 13, we can see that the 6th prime is 13.

What is the 10 001st prime number?

Proof

We find the nth prime with a prime sieve: we cross out the composite numbers up to some limit M, then count the remaining primes in order until we reach the nth one. This only works if M is at least as large as the nth prime, and the sieve does not tell us how large that is in advance. The rest of this proof establishes an upper bound on the nth prime, which we use as the limit M.

Throughout, let p_n denote the nth prime, and for a real number x let $\pi(x)$ denote the number of primes less than or equal to x.

A lower bound on how many primes there are up to x is the same thing as an upper bound on where the nth prime is: if $\pi(x) \geq n$, then at least n primes are at most x, so $p_n \leq x$. It therefore suffices to show that there are many primes up to x. We do this for even x, writing $x = 2m$.

Our approach is to take a specific number N and look at its prime factorization. For a prime p and a positive integer k, let $\nu_p(k)$ denote the exponent of p in the prime factorization of k, that is, the largest e such that $p^e \mid k$. Write

$$N = \prod_{p \leq 2m} p^{\nu_p(N)},$$

with one factor for every prime $p \leq 2m$, using exponent 0 for primes that do not divide N. This product is all of N provided N has no prime factor larger than $2m$, and it has exactly $\pi(2m)$ factors. If, in addition, every factor $p^{\nu_p(N)}$ is at most $2m$, then $N \leq (2m)^{\pi(2m)}$. So if N is very large, $\pi(2m)$ cannot be small: a large product of small factors needs many factors.

We take $N = \binom{2m}{m}$. It is a natural choice because it is large, roughly 4^m, yet it is built only from the numbers $1, 2, \ldots, 2m$, since $\binom{2m}{m} = \frac{(2m)!}{m!\,m!}$. So we expect both its size and its prime factors to be controlled by $2m$. Lemma 1 shows that

it is large, and lemma 4 shows that it has no prime factor larger than $2m$ and that every factor in its prime factorization is at most $2m$.

Lemma 1. *For every integer $m \geq 1$,*

$$\binom{2m}{m} \geq \frac{4^m}{2m+1}.$$

Proof. We begin by applying the binomial theorem to $(1+1)^{2m}$:

$$4^m = 2^{2m} = (1+1)^{2m} = \sum_{k=0}^{2m} \binom{2m}{k} 1^{2m-k} 1^k = \sum_{k=0}^{2m} \binom{2m}{k}.$$

The other observation about binomial coefficients is that the largest value of $\binom{2m}{k}$ is when $k = m$. We show this by analyzing the ratio between consecutive terms $\binom{2m}{k+1}$ and $\binom{2m}{k}$, for $0 \leq k \leq 2m-1$:

$$\frac{\binom{2m}{k+1}}{\binom{2m}{k}} = \frac{\frac{(2m)!}{(k+1)!\,(2m-k-1)!}}{\frac{(2m)!}{k!\,(2m-k)!}} = \frac{k!\,(2m-k)!}{(k+1)!\,(2m-k-1)!}.$$

Since $(k+1)! = (k+1)\cdot k!$ and $(2m-k)! = (2m-k)\cdot(2m-k-1)!$, the factorials $k!$ and $(2m-k-1)!$ cancel, leaving

$$\frac{\binom{2m}{k+1}}{\binom{2m}{k}} = \frac{2m-k}{k+1}.$$

This ratio is greater than 1 exactly when

$$2m-k > k+1 \iff 2k < 2m-1 \iff k < m - \tfrac{1}{2},$$

and since k is an integer, this happens exactly when $k \leq m-1$. In short, when $k < m$ the next term is bigger, and when $k \geq m$ the next term is smaller. So the terms increase up to $k = m$ and decrease after it:

$$\binom{2m}{0} < \binom{2m}{1} < \cdots < \binom{2m}{m} > \binom{2m}{m+1} > \cdots > \binom{2m}{2m},$$

and therefore $\binom{2m}{m}$ is the largest of them.

Since every term of the sum is at most $\binom{2m}{m}$,

$$\sum_{k=0}^{2m} \binom{2m}{k} \leq \sum_{k=0}^{2m} \binom{2m}{m} = (2m+1)\binom{2m}{m}.$$

Combining both results, we get

$$4^m \leq (2m+1)\binom{2m}{m}, \qquad \text{so} \qquad \binom{2m}{m} \geq \frac{4^m}{2m+1}. \qquad \square$$

To bound the factors $p^{\nu_p(N)}$ of $N = \binom{2m}{m}$, we need $\nu_p(N)$. We first show that ν_p turns products into sums. The proof relies on Euclid's lemma: if a prime divides a product of two integers, then it divides at least one of them.

Lemma 2. *For every prime p and all positive integers a and b,*

$$\nu_p(ab) = \nu_p(a) + \nu_p(b).$$

Proof. Let $\alpha = \nu_p(a)$, that is, you can divide a by the prime number p α times. Let a' be the remaining factors when you divide out all the p that fit in a. That is,

$$a = p^\alpha \cdot a'.$$

Similarly, let $\beta = \nu_p(b)$ and b' be the leftover after dividing out all ps that fit in b.

$$b = p^\beta \cdot b'.$$

Multiplying the two we get:

$$ab = p^{\alpha+\beta} \cdot a'b'$$

It follows that $\nu_p(ab) \geq \alpha + \beta$. The only question is if p divides $a'b'$. By construction, p divides neither a' nor b', so by Euclid's lemma p doesn't divide $a'b'$ either. If $p^{\alpha+\beta+1}$ divided ab, then dividing both sides of $ab = p^{\alpha+\beta} \cdot a'b'$ by $p^{\alpha+\beta}$ would show that p divides $a'b'$, which we just ruled out. Therefore, $\nu_p(ab) = \alpha + \beta = \nu_p(a) + \nu_p(b)$. □

Since $\binom{2m}{m} = \frac{(2m)!}{m!\,m!}$, we have $(2m)! = \binom{2m}{m} \cdot m! \cdot m!$. By lemma 2, $\nu_p((2m)!) = \nu_p(\binom{2m}{m}) + 2\,\nu_p(m!)$, that is,

$$\nu_p\left(\binom{2m}{m}\right) = \nu_p((2m)!) - 2\,\nu_p(m!),$$

so it is enough to know how many times p divides a factorial.

Lemma 3 (Legendre's formula)**.** *For every prime p and integer $k \geq 0$,*

$$\nu_p(k!) = \sum_{i=1}^{\infty} \left\lfloor \frac{k}{p^i} \right\rfloor,$$

where the sum is finite because every term with $p^i > k$ is 0.

Proof. Recall that $k!$ is just the product of all positive integers up to and including k. That is $\nu_p(k!) = \nu_p(1 \cdot 2 \cdot 3 \cdots k)$, and by Lemma 2, this expands to:

$$\nu_p(k!) = \sum_{j=1}^{k} \nu_p(j)$$

Next we express $\nu_p(j)$ as a summation, let square brackets denote 1 if the enclosed expression is true and 0 if it is false.

$$\nu_p(j) = \sum_{i=1}^{\infty} \left[p^i \mid j\right]$$

To see why, let $e = \nu_p(j)$, so $j = p^e \cdot j'$ where p does not divide j'. For $i \leq e$, p^i divides p^e, which divides j, so the term is 1. For $i > e$, if p^i divided j then p would divide j at least $i > e$ times, contradicting that e is the largest such count, so the term is 0. The summation is therefore e ones followed by zeros, which is $e = \nu_p(j)$.

Only finitely many terms of this summation are nonzero. Since $p \geq 2$, the binomial theorem gives $p^j \geq 2^j = (1+1)^j \geq 1 + j > j$, so for every $i \geq j$ we have $p^i \geq p^j > j$, and p^i cannot divide j. So every nonzero term has $i < j$.

This leaves us with

$$\nu_p(k!) = \sum_{j=1}^{k} \sum_{i=1}^{\infty} \left[p^i \mid j\right]$$

Since $j \leq k$, every nonzero term has $i < j \leq k$, so every term with $i \geq k$ is 0. Dropping those zero terms leaves a finite sum over $1 \leq j \leq k$ and $1 \leq i \leq k$, and a finite sum may be added in any order. Swapping the order and then restoring the zero terms gives

$$\nu_p(k!) = \sum_{j=1}^{k} \sum_{i=1}^{k} \left[p^i \mid j\right] = \sum_{i=1}^{k} \sum_{j=1}^{k} \left[p^i \mid j\right] = \sum_{i=1}^{\infty} \sum_{j=1}^{k} \left[p^i \mid j\right].$$

We now consider what the term tells us when we fix i and let j vary. The terms of the inner loop now ask which integers $1 \leq j \leq k$ are divided by p^i. Those such integers must then be multiples of p^i. Let $q \geq 0$ be the largest integer such that $qp^i \leq k$. Then the multiples of p^i in $1, \ldots, k$ are exactly $1 \cdot p^i, 2 \cdot p^i, \ldots, q \cdot p^i$, so there are q of them, and

$$\sum_{j=1}^{k} \left[p^i \mid j\right] = q.$$

(If $p^i > k$ there are no multiples, and indeed $q = 0$.)

It remains to identify q. Since $qp^i \leq k$, we have $q \leq \frac{k}{p^i}$. Since q is the largest integer with this property, $q+1$ does not have it, so $(q+1)p^i > k$, that is, $q + 1 > \frac{k}{p^i}$. Together,

$$q \leq \frac{k}{p^i} < q + 1.$$

By definition, $\lfloor x \rfloor$ is the unique integer n with $n \leq x < n + 1$, so $q = \left\lfloor \frac{k}{p^i} \right\rfloor$. Substituting into the swapped sum,

$$\nu_p(k!) = \sum_{i=1}^{\infty} \left\lfloor \frac{k}{p^i} \right\rfloor.$$

□

Applying Lemma 3 to $\nu_p\left(\binom{2m}{m}\right) = \nu_p((2m)!) - 2\,\nu_p(m!)$ we get

$$\begin{aligned}\nu_p\left(\binom{2m}{m}\right) &= \nu_p((2m)!) - 2\,\nu_p(m!) \\ &= \sum_{i=1}^{\infty} \left\lfloor \frac{2m}{p^i} \right\rfloor - 2\sum_{i=1}^{\infty} \left\lfloor \frac{m}{p^i} \right\rfloor \\ &= \sum_{i=1}^{\infty} \left(\left\lfloor \frac{2m}{p^i} \right\rfloor - 2\left\lfloor \frac{m}{p^i} \right\rfloor \right)\end{aligned}$$

Lemma 4. *For every integer $m \geq 1$,*

$$\binom{2m}{m} \leq (2m)^{\pi(2m)}.$$

Proof. TODO □

Together, lemmas 1 and 4 give

$$\frac{4^m}{2m+1} \leq \binom{2m}{m} \leq (2m)^{\pi(2m)}.$$

The left side grows exponentially in m, while the right side can only keep up if the exponent $\pi(2m)$ is large. Taking logarithms makes this precise.

Theorem 5. *For every even integer $x \geq 2$,*

$$\pi(x) \geq \frac{x \ln 2}{\ln x} - 2.$$

Proof. TODO □

Finally, we turn this lower bound on $\pi(x)$ into an upper bound on p_n that we can compute directly, and use it as the sieve limit.

Theorem 6. *For every integer $n \geq 2$,*

$$p_n \leq 4n \ln n.$$

Proof. TODO □